IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-CS-011

1. Topic (12 words max)

DG 17 — Partnerships for the Goals

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The increasing use of **AI in cybercrime** has significantly amplified the effectiveness of **social engineering**, **phishing**, and **malware-based attacks**. Modern scam campaigns now employ **AI-generated videos, audio, and text** that closely resemble legitimate communication, making them **highly persuasive and difficult to detect**. These attacks are frequently distributed through **trusted platforms** such as **social media networks**, **messaging applications**, **QR codes**, and **public code repositories (GitHub, Telegram, etc.)**, which users generally **perceive as safe**. As a result, attackers are able to deceive victims into **voluntarily exposing sensitive information** or **installing malicious software**, leading to **financial and data losses**.

The primary groups affected by this problem are **individual internet users** and **small and medium-sized businesses (SMBs)**. Unlike **large enterprises**, these user groups typically **lack access to enterprise-grade cybersecurity tools** such as **Endpoint Detection and Response (EDR) systems**, **dedicated security teams**, or **continuous monitoring infrastructures**. Existing defenses, including **firewalls**, **antivirus solutions**, and **user awareness training**, are largely **reactive**, **cost-intensive**, or **technically complex**, and are therefore **insufficient for preventing AI-driven scams** that rely on **human interaction rather than direct system exploitation**.

The impact of this problem is **multi-dimensional**. **Socially**, **AI-enhanced scams** exploit **human trust, urgency, and authority**, leading to **psychological manipulation** and **loss of confidence in digital platforms**. **Economically**, victims may suffer **direct financial losses**, **business disruption**, and **reputational damage**. **Technically**, successful attacks can result in **malware infections**, **credential theft**, and **unauthorized access to systems**. **Ethically**, the **misuse of AI technologies** for deception raises serious concerns regarding **responsible AI use** and **digital safety**. These combined impacts highlight the **growing severity and real-world relevance** of the problem.

This research falls under the **Information Assurance & Security research cluster**, with relevance to industry sectors such as **digital platforms**, **financial services**, and **small business IT environments**. Furthermore, the research aligns with the **United Nations Sustainable Development Goals (SDGs)**, particularly:

- **SDG 4 (Quality Education)** – through **user awareness** and **explainable risk indicators**
- **SDG 5 (Gender Equality)** and **SDG 10 (Reduced Inequalities)** – by protecting **vulnerable user groups** disproportionately affected by scams
- **SDG 9 (Industry, Innovation, and Infrastructure)** – by promoting **safer and more resilient digital infrastructure**
- **SDG 8 (Decent Work and Economic Growth)** – by reducing **financial losses** and **operational disruptions** for individuals and SMBs
- **SDG 11 (Sustainable Cities and Communities)** – by strengthening **trust in digital services**
- **SDG 16 (Peace, Justice, and Strong Institutions)** – by contributing to the **prevention of cyber-enabled fraud** and strengthening **trust in digital systems.**
- **SDG 17 (Partnerships for the Goals)** - Encourages collaboration between **academia, industry, CERTs, policymakers**, and platforms

Addressing this problem requires a **lightweight**, **explainable**, and **preventive defense approach tailored specifically to individuals and SMBs** facing **AI-driven scam and malware campaigns**.

[1] Jabir, R., Le, J., & Nguyen, C. (2025). *Phishing Attacks in the Age of Generative Artificial Intelligence: A Systematic Review of Human Factors*. MDPI.

[2] Digital deception: generative artificial intelligence in social engineering and phishing. (*Artificial Intelligence Review*, 57, 2024).

[3] Varela Enríquez, C., Toasa, R., & Urdaneta, M. (2025). *The Use of Artificial Intelligence in Cybercrime: Impact Analysis and Mitigation Strategies*. J. Cybersecur. Priv.

[4] AI-Generated Media Drives Real-World Fraud, Identity Theft, and Business Compromise. *Malay* Mail (2025).

[5] S. K. Singh, R. Kumar, and A. Sharma, "Machine Learning-Based Email Spam Detection System," International Journal of Intelligent Systems and Applications in Engineering (IJISAE), vol. 12, no. 2, pp. 45–53, 2024.

[6] R. Prakash and V. Ramesh, "Detection of AI-Driven Social Engineering Phishing Attacks Using Machine Learning Techniques," PSG College of Arts & Science Institutional Repository, 2023.

[7] M. Alshamrani, A. Alzahrani, and K. Alsubhi, "Adaptive Phishing Detection Framework Using Reinforcement Learning," MDPI Digital Technologies, vol. 5, no. 2, pp. 26–39, 2023.

[8] P. Mehta and S. Patel, "Hybrid Feature Selection and Neural Network Model for Phishing and Spam Detection," International Journal of Engineering Science and Research (IJESR), vol. 14, no. 1, pp. 88–95, 2024.

[9] A. Rahman and J. Kim, "EXPLICATE: Explainable LLM-Enhanced Phishing Detection System," arXiv preprint arXiv:2503.20796, 2025.

6. Brief description of Existing Research and Systems (300 words max)

Existing Products and Systems

Several cybersecurity products and systems currently exist to protect users from malware, phishing, and related cyber threats. Commonly used solutions include **firewalls**, **antivirus software**, **email spam filters**, and **Endpoint Detection and Response (EDR)** systems. These tools are primarily designed to detect known malicious files, suspicious network activity, or abnormal system behavior after an attack has been initiated or executed. Large organizations also employ **Security Information and Event Management (SIEM)** platforms and **managed security services** to monitor and respond to threats in real time.

While these products are effective in enterprise environments, they are generally **resource-intensive**, **costly**, and require **specialized technical expertise** to deploy and maintain. As a result, such solutions are often **inaccessible or impractical** for individual users and small and medium-sized businesses (SMBs). Furthermore, many of these tools focus on technical indicators of compromise and are not designed to address **AI-driven social engineering attacks** that rely on deception, persuasion, and user interaction rather than direct system exploitation.

Existing Research

Existing academic research has extensively studied **traditional phishing attacks**, **malware detection techniques**, and **user awareness training** as separate security domains. Recent studies have begun to explore the role of **artificial intelligence in cybercrime**, particularly the use of AI-generated text, images, audio, and videos to enhance phishing and scam campaigns. Other research has examined **deepfake detection**, **behavioral analysis**, and **platform-based malware distribution** as isolated topics.

However, much of the existing research remains **fragmented**, with individual studies focusing on either technical malware detection, AI-generated media analysis, or human behavioral aspects in isolation. Additionally, many proposed solutions emphasize **highly specialized machine learning models** or **enterprise-scale defenses**, which may not be suitable for real-world deployment by individuals or small organizations. There is limited research that integrates **AI-driven content analysis**, **human-centered behavioral indicators**, and **trusted-platform abuse patterns** into a unified and explainable defensive approach.

Gaps and Limitations Identified

Based on the review of existing systems and research, several key gaps have been identified. First, current products largely adopt a **reactive detection approach**, identifying threats only after system compromise or user interaction has already occurred. Second, existing solutions do not sufficiently address the **human-centric nature** of AI-driven scams, where deception and persuasion are the primary attack vectors. Third, many research efforts lack **practical applicability** for individuals and SMBs due to complexity, cost, or reliance on enterprise infrastructure.

Most importantly, there is a lack of a **lightweight, explainable, and preventive defense framework** that combines insights from **AI-generated scam content**, **social engineering behavior**, and **trusted-platform abuse mechanisms**. This gap highlights the need for a holistic research-driven solution that can provide early risk awareness and actionable guidance to non-technical users before financial, or data loss occurs.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed research presents a practical and evidence-driven solution to address the increasing threat of AI-enhanced social engineering, phishing, and malware campaigns targeting individuals and small and medium-sized businesses (SMBs). Instead of attempting to replace existing cybersecurity tools or claiming full technical prevention, the solution focuses on strengthening the most frequently exploited element in modern attacks: **human interaction with AI-generated deceptive content delivered through trusted digital platforms**.

Current security defenses are largely system-centric and reactive, often detecting threats only after damage has occurred. In contrast, this research emphasizes **early-stage risk identification and user decision support**, enabling users to recognize and avoid malicious interactions before sensitive data is exposed or malware is installed. The solution is designed to be lightweight, explainable, and suitable for environments where enterprise-grade security infrastructure is unavailable.

From a system perspective, the proposed solution operates as follows:

- **Inputs**

  The system observes user-facing digital interactions that commonly appear in AI-driven scam campaigns, including:

  - AI-generated videos, audio, and text used in advertisements, messages, or emails
  - User interactions with links, QR codes, and shared files
  - Contextual information related to delivery platforms such as social media, messaging applications, and public repositories (e.g., Telegram, GitHub)

- **Process**

  The collected inputs are analyzed using a combination of explainable and non-intrusive techniques:

  - Identification of AI-generated scam content characteristics, such as unnatural phrasing, visual inconsistencies, or synthetic speech cues

- Analysis of behavioral manipulation patterns, including urgency, authority, trust exploitation, and emotional pressure
- Evaluation of trusted-platform abuse indicators, such as unusual content delivery paths or misuse of reputable services
- Aggregation of these indicators using transparent and interpretable logic to assess the likelihood of malicious intent, without relying on opaque black-box models

- **Outputs**

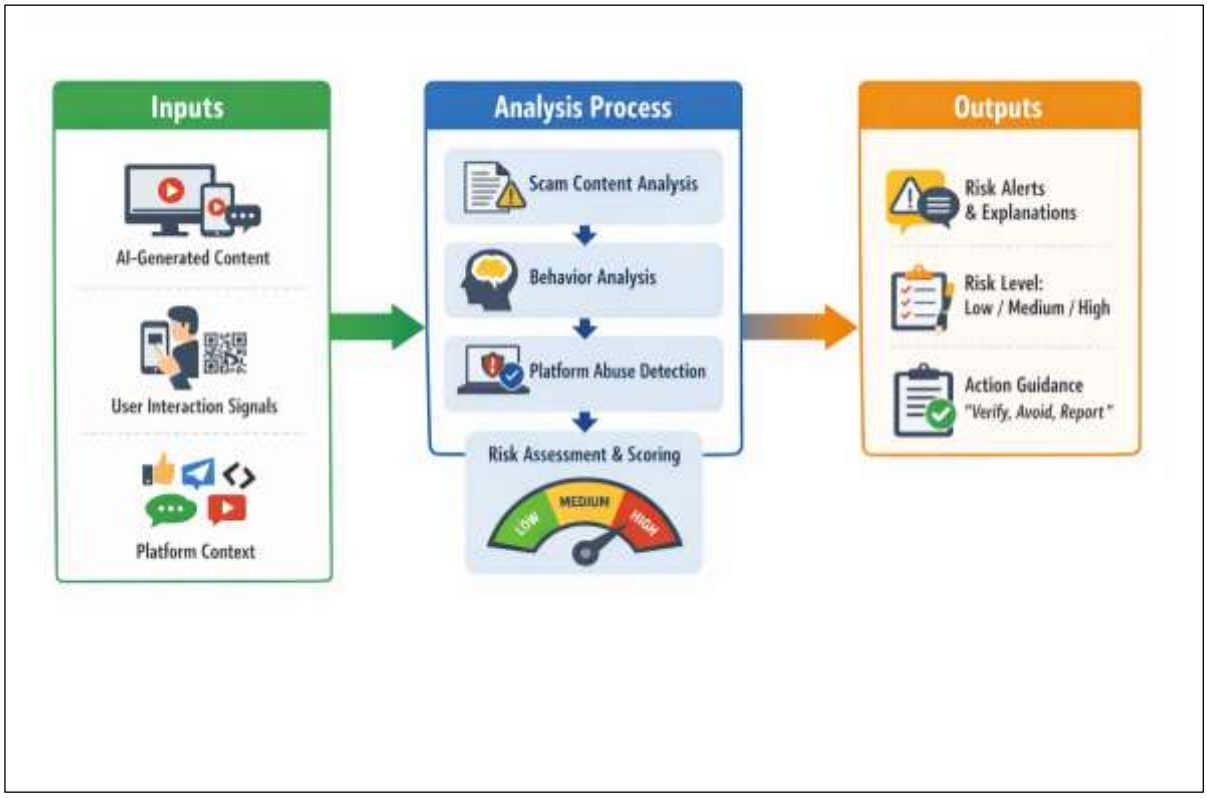
  Based on the assessed risk level, the system provides:

  - Clear and understandable risk warnings at the point of user interaction
  - Explainable justifications describing *why* the content or interaction is considered suspicious
  - Practical guidance to help users decide whether to proceed, verify, or avoid the interaction entirely

The outcome is a **user-centric risk awareness and decision-support model**, rather than a fully automated blocking system. By prioritizing explainability and usability, the solution empowers users to make informed security decisions instead of blindly trusting automated judgments. This approach directly addresses the cost, complexity, and trust limitations of existing defenses, making it particularly suitable for individuals and SMBs.

Overall, the proposed solution delivers a realistic and scalable contribution by integrating technical analysis with human-centric considerations. It strengthens early-stage defense against AI-driven scam and malware campaigns while remaining practical, transparent, and aligned with real-world constraints.

Conceptual Diagram



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

This research requires a combination of **cybersecurity, AI, and human behavioral analysis expertise**, which the project team members will systematically acquire and develop. Team members will gain **essential domain expertise** through a combination of formal coursework in **Cybersecurity, Cryptography, and Artificial Intelligence**, as well as targeted **self-study of recent research papers, industry reports, and case studies** on AI-driven phishing, deepfake scams, and malware campaigns. Participation in online workshops, webinars, and hands-on exercises related to social engineering and malware analysis will further strengthen the team's practical understanding of attack techniques and mitigation strategies.

To obtain the **technical knowledge and skills required**, members will engage in **practical experimentation using AI content detection tools, malware simulation platforms, and threat analysis frameworks**. Skills such as analyzing AI-generated media, evaluating behavioral cues in scam communications, and identifying trusted-platform abuse patterns will be developed through a structured combination of **sandbox testing, tool usage, and guided simulations**. Additionally, members will leverage open-source AI and cybersecurity tools, as well as programming libraries, to implement lightweight analytical models, ensuring the research solution is grounded in practical techniques and real-world applicability.

The **data required for the project** will be collected from multiple sources while ensuring safety, legality, and ethical compliance. Publicly available repositories, social media content, phishing simulation datasets, and AI-generated scam samples from controlled research environments will be used for analysis. All data collection will be carefully curated to exclude personally identifiable information (PII) and any sensitive user data. Controlled datasets and test cases will allow the team to model real-world attack scenarios without exposing participants to actual risk. Data will be documented, categorized, and stored securely to ensure integrity, reproducibility, and ease of analysis throughout the research.

Ethical considerations are critical to this research, particularly because it involves human behavioral modeling and analysis of deceptive AI content. While the research does not involve direct experimentation on human subjects in live attack scenarios, **ethical clearance may still be required** to ensure compliance with institutional guidelines for handling simulated social engineering, behavioral analysis, and synthetic data.

Any human-related observations will use anonymized, simulated, or publicly consented data to comply with ethical and privacy standards.

In summary, the project team has **a clear plan to acquire the necessary expertise**, develop technical skills, collect and curate relevant data safely, and address ethical considerations. These measures ensure that the research is feasible, responsibly conducted, and grounded in both practical and theoretical knowledge, enabling the successful implementation of a lightweight, explainable, and preventive framework for defending individuals and small businesses against AI-driven scams and malware campaigns.

9. Objectives and Novelty

Main Objective *A precise, clear statement that defines what a research project aims to accomplish.*			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Student ID	*Your component of the project*	*Tasks you perform to achieve your component and overall objectives.*	*Your genuine contribution / What you can't observe in other research projects that have tackled the same problem*

10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				